Energy-Efficient Smart Node Selection for Coverage-Aware Redundancy Minimization in IoT-Based Wireless Sensor Networks

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***Abstract*—Wireless Sensor Networks (WSNs) play a fundamental role in Internet of Things (IoT) applications, where sensing efficiency depends on how active sensor nodes are selected and utilized. This paper proposes an energy-efficient smart node selection scheme for IoT-based WSNs that aims to provide high coverage while minimizing redundancy, total energy consumption, and the number of active nodes.**

The problem is formulated as a multi-objective optimization task over a set of candidate nodes distributed in a two-dimensional monitoring field. A heuristic-based approach is introduced to iteratively select feasible active nodes using a node selection score based on coverage gain, redundancy increase, energy cost, and sink connectivity preservation.

Simulation results show that the proposed method achieves a better overall tradeoff than conventional random and grid deployment methods, especially at medium and high node densities, by reducing redundancy, lowering energy consumption, and using fewer active nodes while maintaining full sink connectivity and competitive coverage. These results demonstrate that the proposed node selection strategy provides a practical and efficient solution for redundancy-aware WSN management in IoT environments.

***Index Terms*—Wireless Sensor Networks, Internet of Things, node selection, redundancy reduction, energy efficiency, connectivity preservation**

I. Introduction

Wireless Sensor Networks (WSNs) are widely used in Internet of Things (IoT) applications such as smart cities, industrial monitoring, and environmental sensing. In such applications, sensing performance is strongly influenced by which sensor nodes are activated, since node activation directly affects network coverage, redundancy, energy consumption, and connectivity to the sink.

Conventional deployment methods, such as random and grid-based deployment, often lead to inefficiencies. Random deployment may produce uneven coverage and weak connectivity, while grid deployment can create excessive overlap between sensing regions, leading to redundancy and avoidable energy expenditure. These limitations motivate the need for more intelligent strategies that select only the most useful active nodes from a set of candidates.

This work focuses on the active node selection problem in a two-dimensional monitoring field. Instead of activating all candidate nodes, the proposed scheme selects an energy-efficient subset of active nodes that balances multiple performance objectives, namely coverage, redundancy, energy consumption, and connectivity. A heuristic node selection strategy is developed to evaluate each feasible candidate node according to its contribution to coverage, its redundancy effect, its estimated energy cost, and its ability to preserve sink connectivity.

The proposed approach is evaluated through simulation and compared with random and grid deployment baselines. The results show that the proposed scheme achieves a better overall tradeoff by substantially reducing redundancy and energy use while preserving full connectivity and maintaining competitive coverage.

A. Related Work

Deployment and activation strategies for Wireless Sensor Networks (WSNs) have been extensively studied with a focus on coverage optimization, redundancy reduction, connectivity preservation, and energy efficiency. Coverage remains one of the most important performance metrics in WSNs because it determines how effectively a region of interest is monitored, while also affecting network lifetime and sensing quality [1].

Traditional strategies such as random deployment and grid-based deployment have been widely investigated [2]. Random deployment is simple and practical in inaccessible environments, but it often produces uneven node distribution, coverage holes, and inconsistent connectivity. Grid deployment offers more structured coverage, but it may introduce considerable overlap between sensing regions and therefore increase redundancy and energy waste.

To improve performance, optimization-based approaches such as Particle Swarm Optimization (PSO), Grey Wolf Optimization (GWO), and other metaheuristic methods have been proposed to determine better node configurations and improve coverage and lifetime [3], [5]. Although these methods can achieve strong results, they are often computationally expensive and less suitable for lightweight or real-time scenarios.

Redundancy-aware methods have also been explored. Several works attempt to identify redundant nodes whose sensing regions are already covered and then deactivate or relocate them in order to save energy while maintaining acceptable monitoring performance [1]. This direction is particularly

important in dense WSNs where overlapping sensing areas lead to unnecessary resource consumption.

At the same time, jointly achieving coverage and connectivity remains challenging. Existing studies show that balancing these objectives while limiting node count and energy cost is computationally difficult and often NP-hard [4], [6]. In addition, many existing solutions either emphasize raw coverage improvement or require significant computational overhead, making them less practical for scalable IoT-based WSN management [7].

Motivated by these limitations, this paper proposes a heuristic-based smart node selection scheme that activates only a subset of candidate nodes while explicitly accounting for coverage contribution, redundancy minimization, energy efficiency, and sink connectivity.

B. Paper Contributions

The main contributions of this paper are summarized as follows:

- An energy-efficient smart node selection scheme is proposed for IoT-based WSNs to achieve coverage-aware redundancy minimization while preserving sink connectivity.
- A heuristic node selection strategy is developed in which candidate nodes are scored according to coverage gain, redundancy increase, energy cost, and connectivity feasibility.
- A multi-objective formulation is introduced to jointly consider coverage, redundancy, total energy consumption, and active node count under a target coverage requirement.
- The proposed scheme is evaluated through simulation and compared with random and grid deployment baselines, showing improved overall performance in redundancy, energy efficiency, active-node reduction, and connectivity preservation.

C. Paper Organization

The remainder of this paper is organized as follows. Section II presents the system model. Section III describes the proposed smart node selection scheme, including the mathematical formulation and heuristic algorithm. Section IV discusses the simulation setup and comparative results. Finally, Section V concludes the paper and outlines possible future directions.

II. System Model

The system is an IoT-based Wireless Sensor Network (WSN) deployed over a two-dimensional monitoring field, referred to as the region of interest (ROI). A set of candidate sensor nodes is distributed over the field, while a sink node is placed outside the monitoring area to collect sensed data. Instead of activating all available nodes, the proposed scheme selects only a subset of active nodes in order to achieve high coverage with reduced redundancy, lower energy consumption, and full connectivity to the sink.

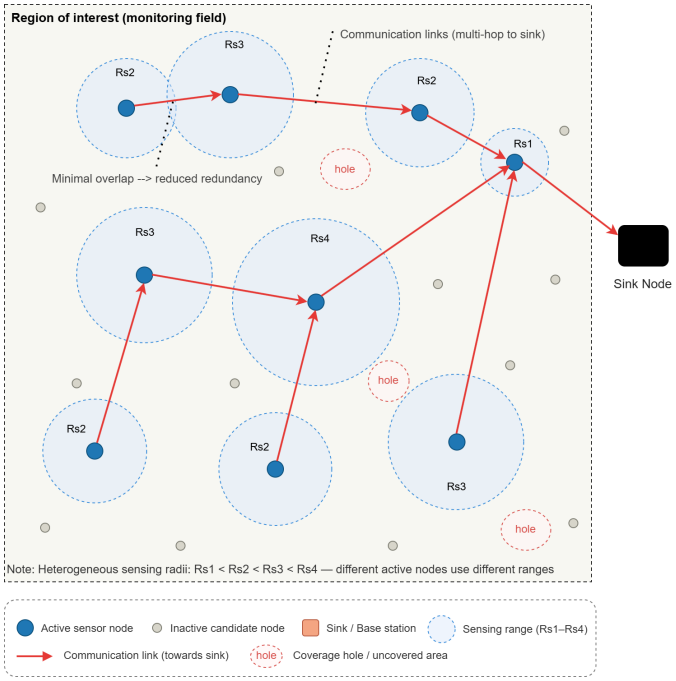

Fig. 1. System model of the proposed IoT-based wireless sensor network with heterogeneous sensing radii and sink connectivity.

The system assumes heterogeneous sensor capabilities, where different candidate nodes may operate with different sensing radii. In particular, each selected active node uses one sensing radius chosen from a predefined set of four possible ranges. In addition, the communication radius of each node is assumed to satisfy the common WSN condition that it is at least twice the sensing radius, which helps preserve network connectivity. To evaluate sensing performance, the monitoring field is discretized into a finite set of target points, and coverage is determined based on whether these points fall within the sensing range of the selected active nodes.

Fig. 1 illustrates the adopted system model. The figure shows the monitoring field, candidate node locations, selected active nodes, heterogeneous sensing ranges, communication links toward the sink, overlap reduction between sensing regions, and small uncovered areas that may remain when redundancy and energy efficiency are prioritized over full coverage.

The proposed system model reflects the main design objective of this work: selecting an energy-efficient set of active nodes that balances coverage, redundancy reduction, and connectivity preservation. This model serves as the basis for the mathematical formulation and heuristic node selection algorithm presented in the next section.

III. Proposed Scheme

The proposed method addresses the active node selection problem in an IoT-based Wireless Sensor Network (WSN). Given a set of candidate sensor nodes, the goal is to select an energy-efficient subset of active nodes that provides high coverage, reduced redundancy, and full connectivity to the sink.

The proposed scheme combines a mathematical formulation with a heuristic node selection strategy.

Proposed Problem Formulation

Consider a two-dimensional monitoring field of size $A = L \times W$ with N candidate sensor nodes.

- Candidate sensor set:

$$\mathcal{S} = \{s_1, s_2, \ldots, s_N\}$$

- Position of node i:

$$s_i = (x_i, y_i)$$

- Binary activation variable:

$$z_i \in \{0, 1\}, \qquad z_i = \begin{cases} 1, & \text{if node } i \text{ is selected as active} \\ 0, & \text{otherwise} \end{cases}$$

- Sensing radius of node i:

$$R_{s,i} \in \{R_{s,1}, R_{s,2}, R_{s,3}, R_{s,4}\}$$

- Communication radius of node i:

$$R_{c,i} \geq 2R_{s,i}$$

- Initial energy of node i:

$$E_i^{init} \in \{E_1, E_2, E_3, E_4\}$$

Let the monitored field be discretized into M target points:

$$\mathcal{P} = \{p_1, p_2, \ldots, p_M\}$$

where each target point is represented as:

$$p_j = (u_j, v_j)$$

Coverage Model

A target point p_j is said to be covered by node s_i if:

$$d_{ij} = \sqrt{(x_i - u_j)^2 + (y_i - v_j)^2} \leq R_{s,i}$$

Define the binary coverage indicator:

$$c_{ij} = \begin{cases} 1, & d_{ij} \leq R_{s,i} \\ 0, & \text{otherwise} \end{cases}$$

A target point is covered if at least one active node covers it:

$$C_j = \min\left(1, \sum_{i=1}^{N} z_i c_{ij}\right)$$

Therefore, the overall coverage ratio is:

$$\text{Cov} = \frac{1}{M} \sum_{j=1}^{M} C_j$$

Redundancy Model

Redundancy occurs when a target point is covered by more than one active node. The redundancy at point p_j is defined as:

$$R_j = \max\left(0, \sum_{i=1}^{N} z_i c_{ij} - 1\right)$$

The normalized redundancy over the whole field is:

$$\text{Red} = \frac{1}{M} \sum_{j=1}^{M} R_j$$

Minimizing Red reduces unnecessary overlap between sensing regions and avoids wasting sensing and communication energy.

Connectivity Model

Two active nodes s_i and s_k can communicate if:

$$d_{ik} = \sqrt{(x_i - x_k)^2 + (y_i - y_k)^2} \leq \min(R_{c,i}, R_{c,k})$$

Define the binary communication indicator:

$$g_{ik} = \begin{cases} 1, & d_{ik} \leq \min(R_{c,i}, R_{c,k}) \\ 0, & \text{otherwise} \end{cases}$$

The selected active nodes must form a connected communication graph such that every active node has at least one communication path to the sink.

Energy Model

The transmission energy for sending a k-bit packet over distance d is:

$$E_{tx}(k, d) = \begin{cases} kE_{elec} + k\epsilon_{fs}d^2, & d < d_0 \\ kE_{elec} + k\epsilon_{mp}d^4, & d \geq d_0 \end{cases}$$

The reception energy is:

$$E_{rx}(k) = kE_{elec}$$

Hence, the total operational energy for all selected active nodes is approximated as:

$$E_{tot} = \sum_{i=1}^{N} z_i \left(E_i^{sense} + E_i^{tx} + E_i^{rx}\right)$$

Active Node Count

The number of selected active nodes is:

$$N_{act} = \sum_{i=1}^{N} z_i$$

Optimization Objective

The proposed problem is formulated as a multi-objective optimization problem:

$$\max \text{ Cov}$$

$$\min \text{ Red}$$

$$\min \ E_{tot}$$

$$\min \ N_{act}$$

These objectives are combined into a single weighted fitness function:

$$\max F = w_1 \cdot \text{Cov} - w_2 \cdot \text{Red} - w_3 \cdot \frac{E_{tot}}{E_{max}} - w_4 \cdot \frac{N_{act}}{N}$$

subject to:

$$\text{Cov} \geq C_{\text{target}}$$

$$\text{Graph}(\{s_i \mid z_i = 1\}) \text{ is connected to the sink}$$

$$z_i \in \{0, 1\}, \qquad \forall i = 1, 2, \ldots, N$$

$$0 \leq x_i \leq L, \qquad 0 \leq y_i \leq W$$

Node Selection Score

At each iteration, the proposed heuristic evaluates every feasible candidate node and assigns it a node selection score. The score of candidate node i is defined as:

$$Q_i = \begin{cases} 1, & \text{if node } i \text{ preserves connectivity to the sink} \\ 0, & \text{otherwise} \end{cases}$$

$$\text{Score}_i = \left(\alpha \cdot \frac{\Delta\text{Cov}_i}{M} - \beta \cdot \frac{\Delta\text{Red}_i}{M} - \gamma \cdot \frac{E_i^{cost}}{E_{\max}} \right) Q_i$$

where:

- ΔCov_i is the number of newly covered target points by candidate node i,
- ΔRed_i is the redundancy increase caused by candidate node i,
- E_i^{cost} is the estimated energy cost of activating candidate node i,
- M is the total number of target points,
- $E_{\max}$ is the maximum node energy cost used for normalization,
- Q_i is the connectivity feasibility indicator,
- α, β, and γ are weighting factors.

In this work, the weights are chosen as:

$$\alpha = 0.6, \qquad \beta = 0.25, \qquad \gamma = 0.15$$

The proposed heuristic then selects the feasible candidate node with the highest score, updates the active-node set, and repeats the process until the target coverage requirement is satisfied.

Input: Candidate node set $\mathcal{S}$, target point set $\mathcal{P}$, sink position, target coverage threshold C_{target}
Output: Selected active-node set $\mathcal{S}_{act}$
Initialize $\mathcal{S}_{act} \leftarrow \emptyset$;
Mark all target points in $\mathcal{P}$ as uncovered;
while *coverage ratio* $\text{Cov} < C_{target}$ *and feasible candidate nodes remain* **do**
  foreach *unselected candidate node* s_i **do**
    Compute the number of newly covered target points, ΔCov_i;
    Compute the redundancy increase, ΔRed_i;
    Estimate the energy cost of activating node s_i, E_i^{cost};
    Check whether node s_i preserves connectivity to the sink;
    if *node* s_i *is feasible* **then**
      Compute the node selection score:
      $$\text{Score}_i = \left(\alpha \cdot \frac{\Delta\text{Cov}_i}{M} - \beta \cdot \frac{\Delta\text{Red}_i}{M} - \gamma \cdot \frac{E_i^{cost}}{E_{\max}} \right) Q_i$$
    end
  end
  Select the feasible candidate node with the highest score;
  Add it to $\mathcal{S}_{act}$;
  Update the covered target points;
end
foreach *node in* $\mathcal{S}_{act}$ **do**
  Temporarily remove the node;
  if *coverage ratio and sink connectivity are preserved* **then**
    Remove the node permanently;
  end
end
Return $\mathcal{S}_{act}$ and the corresponding performance metrics;

Algorithm 1: Coverage-Aware Redundancy-Reduced Smart Node Selection

IV. Simulation Results

A. Simulation Setup

The proposed scheme was evaluated using a Python-based simulation in Google Colab. The network was modeled as a static 100×100 m^2 monitoring field with the sink located at $(110, 50)$ m. The field was divided into a 25×25 target grid, giving $M = 625$ target points. Each candidate node was assigned one heterogeneous sensing radius from $\{12, 16, 20, 24\}$ m, and its communication radius was set to $R_{c,i} = 2R_{s,i}$. The simulation was repeated for 16, 25, 36, 49, and 64 candidate nodes, with 20 independent trials for each case. The main simulation parameters are listed in Table I.

TABLE I
SIMULATION SETUP AND PARAMETER VALUES.

Parameter	Value
Simulation tool	Python / Google Colab
Monitoring field size	100×100 m^2
Sink position	$(110, 50)$ m
Target grid size	25×25 target points
Total target points	$M = 625$
Candidate node counts	16, 25, 36, 49, 64
Sensing radii	$\{12, 16, 20, 24\}$ m
Communication radius	$R_{c,i} = 2R_{s,i}$
Coverage threshold	$C_{\text{target}} = 0.95$
Packet size	$k = 4000$ bits
Electronics energy	$E_{elec} = 50 \times 10^{-9}$ J/bit
Free-space amplifier	$\epsilon_{fs} = 10 \times 10^{-12}$ J/bit/m^2
Multipath amplifier	$\epsilon_{mp} = 0.0013 \times 10^{-12}$ J/bit/m^4
Base sensing energy	$E_{sense}^{base} = 5 \times 10^{-6}$ J
Threshold distance	$d_0 = \sqrt{\epsilon_{fs}/\epsilon_{mp}} \approx 87.71$ m
Node score weights	$\alpha = 0.60$, $\beta = 0.25$, $\gamma = 0.15$
Number of trials	20
Random seed	7

B. Comparison Techniques

The proposed smart node selection scheme is compared against the following benchmark techniques:

- **Random Deployment:** sensor nodes are placed randomly in the monitoring field.
- **Grid Deployment:** sensor nodes are placed uniformly using a regular grid structure.
- **Proposed Smart Node Selection Scheme:** candidate nodes are evaluated and selected using the proposed coverage-aware and redundancy-reduced heuristic.

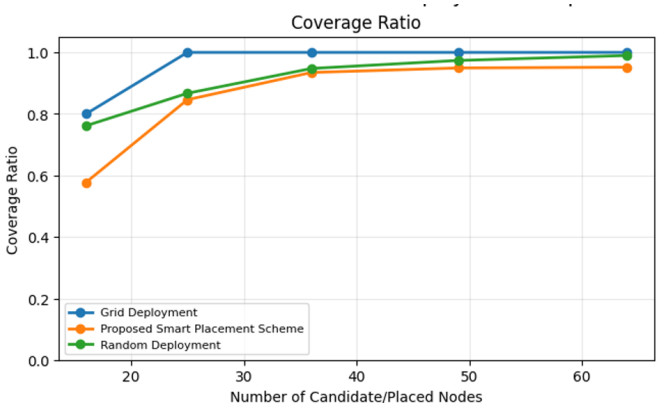

Fig. 2. Coverage ratio versus number of candidate nodes for grid deployment, random deployment, and the proposed smart node selection scheme.

Fig. 2 shows that grid deployment achieves the highest coverage because all nodes are active and uniformly placed. The proposed scheme gives slightly lower coverage, but it remains competitive, reaching 0.952 coverage at 64 candidate nodes.

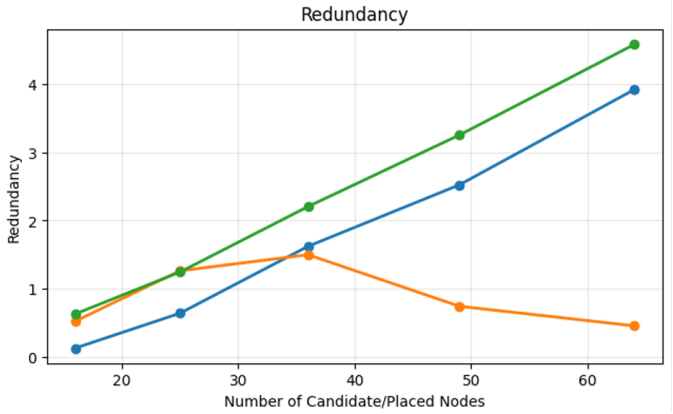

Fig. 3. Redundancy versus number of candidate nodes for different deployment strategies.

Fig. 3 shows that redundancy increases for grid and random deployment as more nodes are activated. The proposed scheme reduces redundancy at higher node counts by selecting only nodes that add useful coverage.

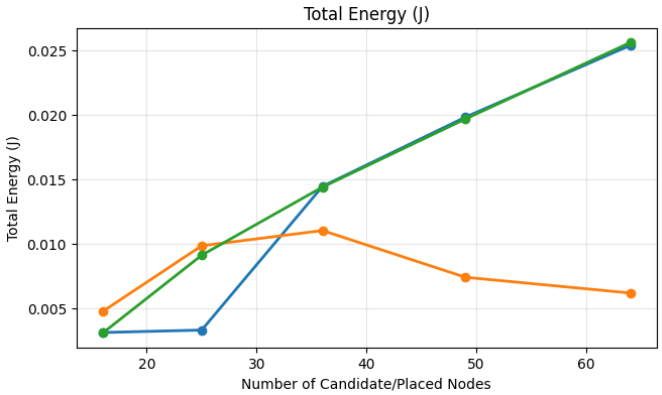

Fig. 4. Total energy consumption versus number of candidate nodes for different deployment methods.

Fig. 4 shows that the proposed scheme becomes more energy-efficient at higher node counts. This is because it activates fewer nodes and avoids unnecessary sensing overlap.

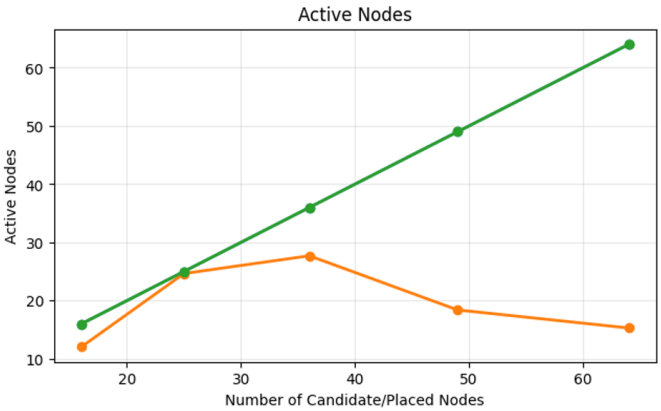

Fig. 5. Number of active nodes selected under different deployment strategies.

Fig. 5 shows that grid and random deployment use all available nodes, while the proposed scheme activates only a selected subset. This reduces the active-node count, especially when more candidate nodes are available.

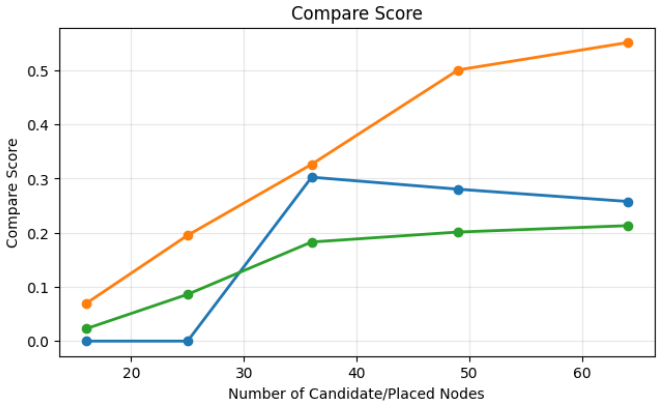

Fig. 6. Comparison score of deployment strategies based on coverage, redundancy, energy efficiency, and connectivity.

Fig. 6 shows that the proposed scheme achieves the highest overall score across all tested node counts. This indicates a better balance between coverage, redundancy, energy consumption, and connectivity.

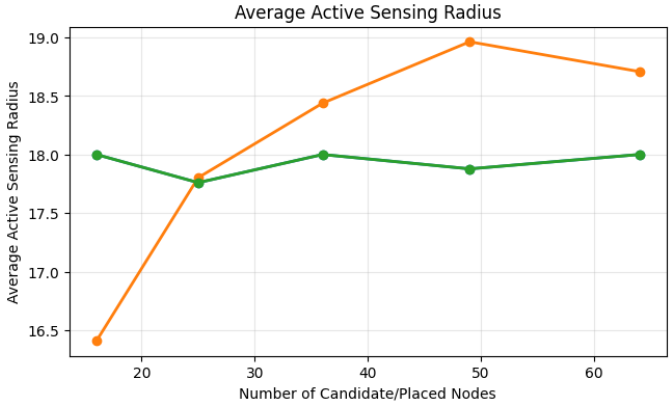

Fig. 7. Average active sensing radius for different deployment methods.

Fig. 7 shows the average sensing radius of the active nodes selected by each method. The proposed scheme may select different sensing ranges depending on which nodes provide the best coverage and connectivity tradeoff.

C. Summary Table

Table II summarizes the average performance of the proposed smart node selection scheme, random deployment, and grid deployment for different node counts. The comparison is reported in terms of coverage ratio, redundancy, total energy consumption, number of active nodes, connectivity, average active sensing radius, node score, compare score, and final rank.

D. Summary of Simulation Results

As shown in Table II, the proposed scheme achieves the best overall rank across all tested node counts. Although grid deployment sometimes attains higher raw coverage, the proposed method provides substantially lower redundancy, lower energy consumption, fewer active nodes, and full sink connectivity, resulting in a better overall tradeoff.

More specifically, the results indicate that the proposed scheme is especially effective at medium and high node counts, where redundancy reduction becomes increasingly important. While the grid method often achieves the largest coverage values, it does so at the cost of significantly higher redundancy and a much larger number of active nodes. In contrast, the proposed scheme activates only the nodes that provide meaningful coverage contribution while limiting unnecessary overlap.

The energy results also show that the proposed strategy maintains a lower operational cost than the baseline methods across most tested cases. At the same time, the connectivity metric confirms that the selected active-node set remains fully connected to the sink, which is a necessary requirement for practical WSN operation. These results demonstrate that the proposed method does not aim to maximize raw coverage alone, but rather to achieve a balanced and energy-efficient node selection solution with strong overall network performance.

TABLE II
COMPARISON OF DEPLOYMENT METHODS FOR DIFFERENT NODE COUNTS USING HETEROGENEOUS SENSING RADII.

Method	**Nodes**	**Cov**	**Red**	E_{tot}	N_{act}	**Conn.**	**Avg.** R_s	**Compare Score**	**Rank**
Grid Deployment	16	0.800	0.128	0.00315	16.00	0.00	18.00	0.0000	3
Proposed Smart Node Selection Scheme	16	0.578	0.523	0.00479	12.05	1.00	16.41	0.0695	1
Random Deployment	16	0.762	0.629	0.00311	16.00	0.20	18.00	0.0228	2
Grid Deployment	25	1.000	0.640	0.00333	25.00	0.00	17.76	0.0000	3
Proposed Smart Node Selection Scheme	25	0.846	1.263	0.00986	24.65	1.00	17.80	0.1952	1
Random Deployment	25	0.867	1.245	0.00914	25.00	0.45	17.76	0.0863	2
Grid Deployment	36	1.000	1.622	0.01447	36.00	1.00	18.00	0.3024	2
Proposed Smart Node Selection Scheme	36	0.935	1.499	0.01105	27.70	1.00	18.44	0.3262	1
Random Deployment	36	0.948	2.206	0.01441	36.00	0.90	18.00	0.1828	3
Grid Deployment	49	1.000	2.525	0.01983	49.00	1.00	17.88	0.2803	2
Proposed Smart Node Selection Scheme	49	0.949	0.743	0.00743	18.40	1.00	18.96	0.5004	1
Random Deployment	49	0.974	3.256	0.01969	49.00	0.95	17.88	0.2013	3
Grid Deployment	64	1.000	3.922	0.02538	64.00	1.00	18.00	0.2577	2
Proposed Smart Node Selection Scheme	64	0.952	0.456	0.00621	15.30	1.00	18.71	0.5508	1
Random Deployment	64	0.990	4.583	0.02560	64.00	1.00	18.00	0.2130	3

V. CONCLUSION

This paper presented an energy-efficient smart node selection strategy for Wireless Sensor Networks in Internet of Things applications. The proposed method addresses key challenges in WSN management by jointly considering coverage, redundancy, energy consumption, active-node count, and sink connectivity.

A heuristic-based node selection algorithm was developed to evaluate candidate nodes according to their coverage gain, redundancy increase, energy cost, and connectivity feasibility. Simulation results show that the proposed method provides a better overall tradeoff than random and grid deployment baselines, especially by reducing redundancy, lowering total energy consumption at higher node densities, and limiting the number of active nodes while preserving full sink connectivity.

Overall, the proposed approach provides a practical and effective solution for coverage-aware redundancy minimization in IoT-based WSNs. One remaining limitation is that the current evaluation assumes a static monitoring field with fixed node positions, fixed traffic behavior, and no node failures. Future work may extend the model to dynamic environments, adaptive traffic conditions, node failure scenarios, or more advanced optimization strategies.

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